3NF — Before

```
CREATE TABLE Student_3NF_Before (
   StudentID INT PRIMARY KEY,
   StudentName VARCHAR(50),
   DepartmentID VARCHAR(10),
   DepartmentName VARCHAR(100)
);

INSERT INTO Student_3NF_Before VALUES
(101, 'Amit', 'D01', 'Computer Science'),
(102, 'Neha', 'D02', 'Mathematics'),
(103, 'Ravi', 'D01', 'Computer Science');
```

3NF — After

```
CREATE TABLE Student_3NF (
   StudentID INT PRIMARY KEY,
   StudentName VARCHAR(50),
   DepartmentID VARCHAR(10)
);

INSERT INTO Student_3NF VALUES
(101, 'Amit', 'D01'),
(102, 'Neha', 'D02'),
(103, 'Ravi', 'D01');


CREATE TABLE Department (
   DepartmentID VARCHAR(10) PRIMARY KEY,
   DepartmentName VARCHAR(100)
);

INSERT INTO Department VALUES
('D01', 'Computer Science'),
('D02', 'Mathematics');
```

BCNF — Before

Use a relation where two independent determinants create a BCNF violation:

```
CREATE TABLE StudentCourseInstructor (
```

```
    StudentID INT,
    CourseID VARCHAR(10),
    Instructor VARCHAR(50),
    PRIMARY KEY (StudentID, CourseID)
);

INSERT INTO StudentCourseInstructor VALUES
(101, 'C01', 'Dr. Rao'),
(102, 'C01', 'Dr. Rao'),
(101, 'C02', 'Dr. Shah'),
(103, 'C02', 'Dr. Shah');
```

Assume:

text
Instructor → Course
(StudentID, CourseID) → Instructor

Decompose:

```
CREATE TABLE InstructorCourse (
    Instructor VARCHAR(50) PRIMARY KEY,
    CourseID VARCHAR(10)
);

INSERT INTO InstructorCourse VALUES
('Dr. Rao', 'C01'),
('Dr. Shah', 'C02');


CREATE TABLE StudentInstructor (
    StudentID INT,
    Instructor VARCHAR(50),
    PRIMARY KEY (StudentID, Instructor)
);

INSERT INTO StudentInstructor VALUES
(101, 'Dr. Rao'),
(102, 'Dr. Rao'),
(101, 'Dr. Shah'),
(103, 'Dr. Shah');
```

4NF — Before

Demonstrate two independent multivalued facts about a student:

```
CREATE TABLE StudentSkillLanguage (
   StudentID INT,
   Skill VARCHAR(50),
   Language VARCHAR(50)
);

INSERT INTO StudentSkillLanguage VALUES
(101, 'Java', 'English'),
(101, 'Java', 'Hindi'),
(101, 'Python', 'English'),
(101, 'Python', 'Hindi');
```

This creates combinations of independent multivalued attributes.

4NF — After

```
CREATE TABLE StudentSkill (
   StudentID INT,
   Skill VARCHAR(50),
   PRIMARY KEY (StudentID, Skill)
);

INSERT INTO StudentSkill VALUES
(101, 'Java'),
(101, 'Python');


CREATE TABLE StudentLanguage (
   StudentID INT,
   Language VARCHAR(50),
   PRIMARY KEY (StudentID, Language)
);

INSERT INTO StudentLanguage VALUES
(101, 'English'),
(101, 'Hindi');
```

5NF — Before

Use three independent relationships:

```
CREATE TABLE StudentCourseTeacher (
   StudentID INT,
   CourseID VARCHAR(10),
   TeacherID VARCHAR(10)
);

INSERT INTO StudentCourseTeacher VALUES
(101, 'C01', 'T01'),
(101, 'C01', 'T02'),
(101, 'C02', 'T01'),
(102, 'C01', 'T01');
```

5NF — After

Decompose into the three pairwise relations:

```
CREATE TABLE StudentCourse (
   StudentID INT,
   CourseID VARCHAR(10),
   PRIMARY KEY (StudentID, CourseID)
);

INSERT INTO StudentCourse VALUES
(101, 'C01'),
(101, 'C02'),
(102, 'C01');


CREATE TABLE StudentTeacher (
   StudentID INT,
   TeacherID VARCHAR(10),
   PRIMARY KEY (StudentID, TeacherID)
);

INSERT INTO StudentTeacher VALUES
(101, 'T01'),
(101, 'T02'),
(102, 'T01');
```

```
CREATE TABLE CourseTeacher (
    CourseID VARCHAR(10),
    TeacherID VARCHAR(10),
    PRIMARY KEY (CourseID, TeacherID)
);

INSERT INTO CourseTeacher VALUES
('C01', 'T01'),
('C01', 'T02'),
('C02', 'T01');
```

These examples should be treated as dependency demonstrations, not as claims that every relation with this shape must be decomposed. For 4NF and especially 5NF, the decomposition depends on the actual multivalued and join dependencies of the relation.